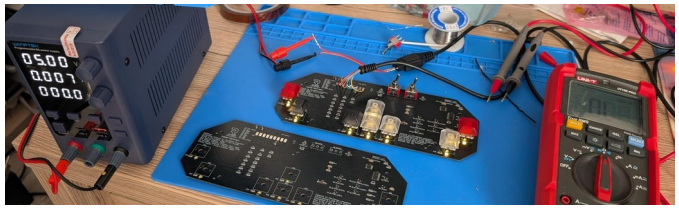
— HARDWARE AND ERGONOMICS

The enclosure, keys and wrist rest are all designed for usability

Together, the electronics and mechanical design give the controller its distinctive character.



***Rev1.5.2** board — the first, still hand-built units undergoing functional testing.*

Eight **Gateron KS-8 Yellow** mechanical switches sit in Kailh hot-swap sockets, allowing the feel of the keys to be customised later. A CMOS 555 timer and CD4066 switching circuit generate **FIRE1** autofire at two selectable speeds. Resettable overcurrent protection and ESD/TVS components provide carefully engineered electrical protection suitable for the original computers.

While I worked on the schematic, PCB and logic, Dester3D — Sándor Sebesi — kept refining the enclosure through six iterations until we achieved the result we wanted. The two-piece shell can be made from PLA materials in different colours; it holds the PCB precisely, snaps together and can also be secured with five M3 screws when needed. Rubber feet support it underneath, while a dedicated strain-relief clip keeps accidental tension on the DB9 cable away from the solder joints.

We also developed a separate, soft black **TPU 85A** wrist rest for greater comfort during longer gaming sessions. It is not just a cushion placed in front of the controller: it connects to the enclosure at three points, so the tested unit stays stable without rocking or sliding. Its fit and comfort have already been checked, while long-term durability testing is still ongoing.

— DEVELOPMENT AND COMMUNITY

From workshop tests to community play in Miskolc

*Alongside the prototypes, player feedback has also helped shape **WASDPad+**.*

After Rev1.0 implemented the core functions, Rev1.2 demonstrated that a joystick built around mechanical keys really works. Rev1.5.1 already featured hot-swap sockets, a second FIRE signal, electrical protection and an LED 'halo' indicator. I subsequently refined the logic, timing and additional features in many areas, with an approach that also accounts for automated manufacturing. [Yes, the **WASDPad+ Rev1.5.2** board is ready for series production — and it needs to be, because we have passed 100 individual components per board. That's the price of discrete logic.] I manually corrected a wiring error in the **SPEED** switch's component footprint on four early boards, and all four passed testing. The new **Rev1.5.2** PCB incorporates the correction into the design and has passed every functional test as well as the extensive design and manufacturing criteria communicated to us by Commodore.

At the public Commodore Reunion test in Miskolc, we held competitions using a four-player joystick splitter, including the crowd favourite Straight Up! and the four-player multiplayer mode of Wizard of Wor. The fast-paced, competitive sessions exceeded all our expectations in terms of community feedback: several players wanted their own **WASDPad+** immediately, and a substantial pre-order queue formed.

But the story does not end there. Hungarian developers are already working on new C64 games built around two-button controls, while others are planning 'WASDPad Edition' adaptations of earlier titles with modified gameplay mechanics. I am also porting a pinball game previously available only for DOS/PS1 to the Amiga, with **WASDPad+** support. Meanwhile, talks are underway with Commodore International about official Commodore-approved status for the electronics. We have already received formal permission to use the classic Commodore emblem on the enclosure and have sent sample units for further evaluation. This is not yet general product approval, however, so we are not using the Commodore logo for the time being.

SMALL BATCH / 2026

WASDPad+

We're returning to the world of retro gaming with mechanical keys.



*The very first Rev1.2 **WASDPad+** development units.*

A lightning-fast mechanical WASD controller for competitive play on the C64 and Amiga, featuring two independent FIRE lines, hardware autofire with selectable speed, interchangeable key sets, LED backlighting, ESD protection, discrete logic, and a symmetrical layout for left- and right-handed players.

The story continues through the games.

Early units are going to developers and testers so that real gameplay can inspire new features, C64 games and WASDPad Edition adaptations. Thank you for supporting our work!

github.com/StingerHU/WASDPad

Electronics: StingerHU / Enclosure: Dester3D – Sándor Sebesi

The images are original photographs and project renders, only cropped and resized.

— HOW THE DESIGN TOOK SHAPE

A real joystick with familiar mechanical keys

Designed for original C64 and Amiga hardware, but delivering an entirely different playing experience.

WASDPad+ is not a USB adapter or a keyboard emulator. It sends native digital direction and FIRE signals straight to the computer through the classic nine-pin Atari-style joystick connector. There is no need for a potentially lag-inducing software-driven microcontroller, firmware or drivers: discrete-logic electronics handle every mechanical keypress entirely in hardware.

Each of the four directions has its own switch, so rapid changes of direction and even simultaneous keypresses feel quite different from moving a joystick. **FIRE1** and **FIRE2** are both available on either side, making the controller equally suitable for left- and right-handed players without switching modes or modifying the hardware. The symmetrical design is a deliberate ergonomic choice, not decoration.



The symmetrical button layout accommodates either hand.

The fast, precise mechanical switches and two independent FIRE signals offer new ways to control familiar games, as well as a fresh playing experience. **WASDPad+** works with the C64, Amiga and other digital joystick ports that use an electrically compatible pinout; the DB9 connector shape alone does not guarantee compatibility.

— USING THE CONTROLLER

How the two fire buttons and hardware autofire work

***WASDPad+** is ready to use straight away: all you need to know is which port to use and how to set the switches.*

DIRECTIONS: ↑ ← ↓ → FIRE1 × 2 / FIRE2 × 2

Switch off your C64 or Amiga before use, then connect the controller to a compatible digital joystick port appropriate for the game. Many titles use port 2, although this depends on the particular software. The four individual mechanical keys provide the directions, while **FIRE1** works as a conventional fire button when **AUTO** is switched off.

FIRE2 is a separate second control signal, used only by computers and games that support it. Two physical buttons are connected to each FIRE line, so switching between left- and right-handed play requires no adjustment. The buttons on either side send the same corresponding FIRE signal.

With **AUTO** switched on, the controller's own circuitry generates repeated fire pulses while **FIRE1** is held down. Move the **SPEED** selector left for the slower SLOW setting or right for the faster FAST setting. A separate power indicator shows that the unit is receiving power, a two-colour LED displays autofire status, and the eight warm-white key backlights can be switched independently using the switch on the underside of the board.

The Gateron mechanical switches are mounted in Kailh MX hot-swap sockets, allowing them to be replaced without soldering using a suitable switch puller (or even by hand). Always power off the computer before changing switches. Never connect the controller to a PC serial port or an old DB9 video port.

— REAL-WORLD GAME TESTS

We tested it with pinball games, action titles and two-button classics

Testing went far beyond a single game or a single computer.

Most C64 and Amiga games designed for standard digital joysticks can be played with **WASDPad+** without modification. **Turrican 3** showed just how different the same game can feel with fast, responsive mechanical buttons; separating jumping and firing onto different buttons brings a completely different experience to these classic platforms.

On the Amiga 1200, we also put the controller through its paces with numerous pinball games besides **Pinball Dreams**. Our two-button tests included Apidya, Street Fighter II, Lionheart and Aladdin; we also examined how the features work in Chaos Engine and Alien Breed using **WHDLoad**. Support for the second FIRE signal always depends on the particular game and its configuration. [Note: with **WHDLoad**, two-button control works smoothly for around 90% of the game library where the CD32 control scheme already provides it by default.]

In an appropriate **WHDLoad** Slave version of **Pinball Dreams**, for example, the 'Full control by 2 joysticks' mode allows the left and right directional keys to operate both flippers, even at the same time. Enter the following tested options in the game's launch icon under Tool Types:

```
SLAVE=PinballDreams.Slave
PRELOAD
CUSTOM2=1
```

CUSTOM2=1 is not a general WHDLoad setting: its meaning may vary between games and Slave versions. Older installations may also need a separate DATA=PD_DATA line. WASDPad+ sends conventional digital joystick signals; it does not emulate a complete CD32 joypad.



We also gathered hands-on experience and feedback at public events during testing.